

ARNAB BHATTACHARYYA

Gopalpur, West Bengal, India

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SUMMARY

AI/ML Engineer with a B.Tech in Artificial Intelligence & Machine Learning and hands-on experience building RAG pipelines, NLP text-processing workflows, and trained classification models using Python, LangChain, and Scikit-learn, alongside production backend/API development in Node.js and Flask.

EDUCATION

Brainware University

B.Tech in CSE (Artificial Intelligence & Machine Learning) — CGPA: 9.53

Aug 2021 – July 2025

Barasat, West Bengal

EXPERIENCE

Capsitech IT Solution

Assistant System Engineer (Software Development) – L1

July 2025 – June 2026

Jodhpur, Rajasthan – On-site

- Engineered **10+ full-stack features** using **TypeScript** across **Node.js** and **React/Next.js**, building **REST APIs** and optimizing data workflows with **MongoDB** for improved query performance and reduced latency.
- Developed **SEO-optimized UI components** for **CapsiCore**, improving crawlability via **structured metadata**, routing strategies, and form handling using **Zod** and **Formik**.

Aeka Advisors

Web Developer Intern

April 2025 – June 2025

Kolkata, West Bengal – On-site

- Built a 10+ component **SvelteKit** UI system, reducing duplication and improving dev speed.
- Centralized application state with **Svelte stores**, removing prop drilling across 5+ shared UI components.

Cehpoint E-Learning & Cyber Security Solutions

MERN Stack & AI Integration Intern

Oct 2024 – Feb 2025

Remote

- Contributed to full-stack development using the **MERN stack** (**MongoDB**, **Express.js**, **React.js**, **Node.js**).
- Enabled secure authentication and real-time database capabilities via **Firebase**, validated with 100+ users.

PROJECTS

ContractLens – Intel® Unnati Industrial Training | Python, Scikit-learn, spaCy, NLTK, Flask

- Built an **NLP pipeline** for legal contract analysis: **PyMuPDF**-based text extraction, preprocessing (lemmatization, stopword removal) with **spaCy** and **NLTK**, and **TF-IDF** vectorization.
- Trained and combined classification models (**MLPClassifier**, **SVC**, **Logistic Regression** via a **VotingClassifier** ensemble) for automated legal clause detection on a labeled clause dataset.
- Implemented cosine-similarity scoring and diff-based highlighting to compare contract clauses, exposed via a **Flask REST API**.

Talk-to-PDF – Python, LangChain, ChromaDB, Sentence-Transformers

- Built a **RAG-based** CLI application for querying PDF documents using natural language.
- Implemented document chunking, embeddings, and semantic retrieval using **LangChain** and **ChromaDB**.
- Generated source-aware responses with retrieved context and similarity-based search.

Groot – TensorFlow, Keras, Flask, Next.js

- Trained a **CNN** for plant leaf disease classification using **TensorFlow/Keras**, achieving **99% training** and **97% validation accuracy** across multiple disease categories.
- Exposed the trained model as a **Flask REST API** for image-based inference, returning disease diagnosis and management guidance.
- Built a **Next.js** web client for image upload and instant diagnosis display.

SKILLS

AI/ML & NLP LangChain, Retrieval-Augmented Generation (RAG), Vector Databases, NLP, Scikit-learn, NLTK, SpaCy, NumPy, Pandas, TensorFlow, Keras, PyTorch

Languages Python, JavaScript, TypeScript, C

Technologies Flask, React.js, Next.js, SvelteKit, Node.js, Express.js, REST APIs, JWT Authentication, Zod, Formik

Databases MongoDB, PostgreSQL, MySQL

DevOps & Cloud Git, GitHub, GitHub Actions (CI/CD), Docker, Vercel, Render

AWARDS & CERTIFICATIONS

- Top 10, HustleX – IIM Lucknow [Certificate] | Finalist, Ideathon – IIM Bangalore [Certificate]
- Intel® Unnati Industrial Training – Business Contract Validation (May – Jul 2024) [Certificate]